

- If key **MTop** is present, it SHALL have a value of "_" (i.e. **MTop=_**). This is the "onboarding payload placeholder".
- Any key whose name begins with **MT** not mentioned in the previous bullets SHALL be reserved for future use by this specification. Manufacturers SHALL NOT include query keys starting with **MT** in either the **CommissioningCustomFlowUrl** or **CallbackUrl** unless they are referenced by a version of this specification.

When the **CommissioningCustomFlowUrl** for a Custom Commissioning Flow device includes the **MTop** key, the **Passcode** embedded in any Onboarding Payload placed on-device or in packaging SHALL NOT be one that can be used for secure channel establishment with the device. This requirement is intended to ensure a shared secret used for proof of possession will not be transferred to a server without user consent. A Custom Commissioning Flow device MAY utilize Onboarding Payload fields such as the Serial Number (see **kTag_SerialNumber**) to pass device identification to the server specified in **CommissioningCustomFlowUrl**, as these fields by themselves could not be used to gain access to the device on their own like the Passcode could.

When the **CommissioningCustomFlowUrl** for a Custom Commissioning Flow device includes the **MTop** key, the **Passcode** embedded in any Onboarding Payload placed on-device or in packaging MAY be set to 0 (one of the **invalid values**) in order to provide a hint to the Commissioner that it is not one that can be used for secure channel establishment with the device. This would allow the Commissioner to avoid attempting to commission the device if an advertisement from it is detected.

Any other element in the **CommissioningCustomFlowUrl** query field not covered by the above rules, as well as the fragment field (if present), SHALL remain as obtained from the **Distributed Compliance Ledger's** **CommissioningCustomFlowUrl** field, including the order of query key/value pairs present.

5.7.3.1.1. Expansion of CommissioningCustomFlowUrl by Commissioner

Once the URL is obtained, it SHALL be expanded to form a final URL (**ExpandedCommissioningCustomFlowUrl**) by proceeding with the following substitution algorithm on the original **CommissioningCustomFlowUrl**:

1. If key **MTcu** is present, compute the **CallbackUrl** desired (see [Section 5.7.3.2, "CallbackUrl format for Custom Commissioning Flow response"](#)), and substitute the placeholder value "_" (i.e. in **MTcu=_**) in the **CommissioningCustomFlowUrl** with the desired contents, encoded with proper URL-encoded escaping (see [RFC 3986](#) section 2).
2. If key **MTop** is present, substitute the placeholder value "_" (i.e. in **MTop=_**) in the **CommissioningCustomFlowUrl** with either **Manual Pairing Code**, or **QR code body** including the **MT**: prefix and TLV data (if present), encoded with proper URL-encoded escaping (see [RFC 3986](#) section 2). Note that both methods SHOULD be supported by the Manufacturer's custom flow.

A Commissioner SHALL NOT append the **MTop=** query key/value pair unless the key/value pair was already present as **MTop=_** in the **CommissioningCustomFlowUrl** previously obtained. This constraint enables the determination of which products make use of the payload in their Custom Commissioning Flow infrastructure by inspection of the Distributed Compliance Ledger records.

The final URL after expansion (**ExpandedCommissioningCustomFlowUrl**) SHALL be the one to follow per [Section 5.7.3, "Custom Commissioning Flow"](#), rather than the original value obtained from the Distributed Compliance Ledger.